

PhishGuard

Hybrid ML + OSINT Architecture for
Proactive Phishing Detection

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BSc Thesis Defense · June 2026

The Problem

Phishing is accelerating

- **\$10.3 billion** annual losses (FBI)
- **1.2M+** unique phishing sites in 2023
- **80%+** of phishing sites now use HTTPS

Blacklists are too slow

- Must be **reported & verified** first
- **50%** of phishing domains last < **12 hours**
- Zero-day attacks hit *before* blacklists update

Existing ML models are context-blind

A URL like `secure-login-chase-update.com` looks structurally normal. Only live intelligence reveals: **registered 2 days ago, no email configured, privacy-protected.**

Our Approach

Combine three intelligence layers into one explainable verdict.

Machine Learning

XGBoost classifier
21 features
96.5% accuracy
0.1ms inference

OSINT

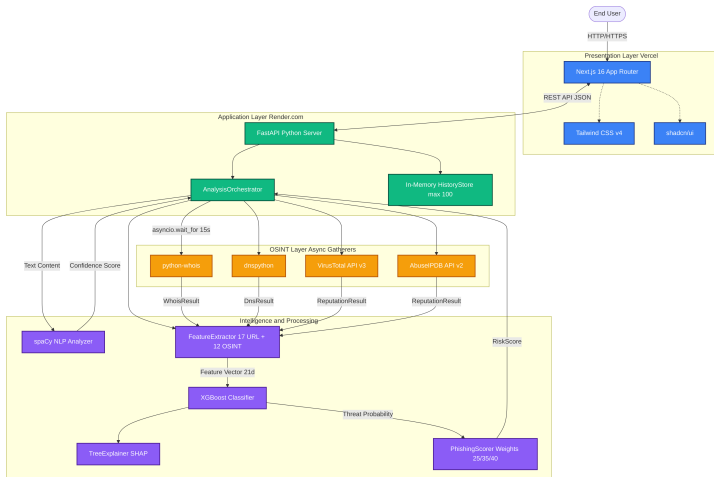
Live WHOIS & DNS
VirusTotal & AbuseIPDB
15s parallel timeout
Graceful fallback

NLP Analysis

spaCy PhraseMatcher
10 tactic families
120+ phrase patterns
Fully explainable

→ **Verdict** (Safe · Suspicious · Dangerous · Critical) with confidence score & human-readable reasons

System Architecture



How do we train the model?

Data Pipeline



Final test set: 5,009 samples — perfectly balanced

2,505 legitimate | 2,504 phishing

21 Features — 17 Structural + 4 OSINT

17 URL Structural Features

urlLength	total URL length
domainLength	domain name length
subdomainCount	subdomain levels
pathDepth	path segments
hasIpAddress	IP as domain
hasAtSymbol	@ in URL
hasDoubleSlash	// in path
hasDashInDomain	hyphen in domain
hasUnderscoreInDomain	
isHttps	HTTPS used?
hasPortNumber	non-std port
hasSuspiciousTld	.tk, .ml, .xyz
hasEncodedChars	% encoding
hasSuspiciousKeywords	
digitRatio	
specialCharCount	
queryParamCount	

4 OSINT Features

hasValidMx	mail configured?
usesCdn	CDN masking?
dnsRecordCount	DNS footprint
hasValidDns	domain resolves?

Key innovation

OSINT features live **inside** the ML vector — the model learns from *real-time Internet state*, not just static URL strings.

Model Performance

Metrics — Test Set (5,009)

Accuracy 96.45%

Precision 97.86%

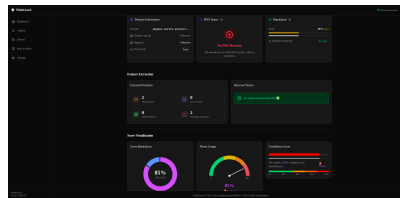
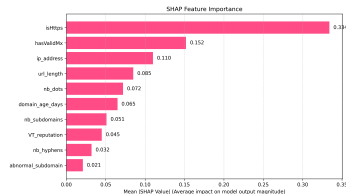
Recall 94.97%

F1-Score 96.39%

AUC-ROC 99.41%

Optuna — Best Trial #43

max_depth 7
learning_rate 0.177
n_estimators 700



How do we know *why* it made that decision?

SHAP Explainability

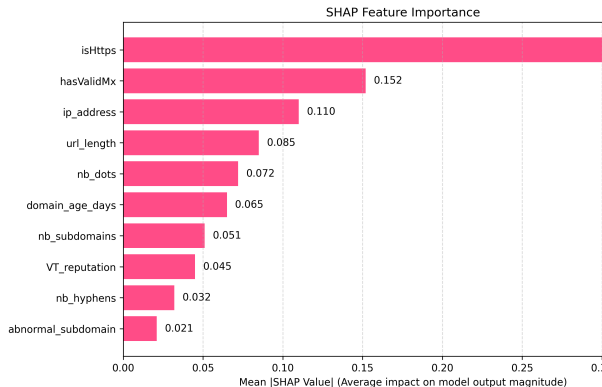
What SHAP does

For every prediction, SHAP calculates each feature's **marginal contribution**:

- “hasValidMx=False pushed the score +0.15 toward phishing”
- “isHttps=True pushed the score -0.08 toward safe”

OSINT matters

14.36% of total SHAP importance comes from the 4 OSINT features alone.



OSINT: Live Internet Intelligence

OSINT — Three Parallel Lookups

WHOIS

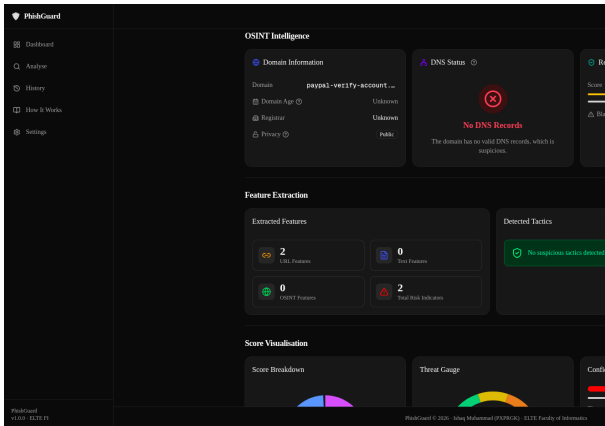
- Domain age, registrar, expiration
- Privacy protection detection
- Red flag:** Registered < 30 days ago

DNS

- A, AAAA, MX, NS, TXT, CNAME
- CDN detection (Cloudflare, Akamai, ...)
- Red flag:** No MX records configured

Reputation

- VirusTotal (70+ engines)
- AbuseIPDB



NLP: Detecting Social Engineering

NLP — 10 Tactic Detectors

spaCy PhraseMatcher — Rule-Based

1. **Urgency** — *“act now”, “24 hours”*
2. **Threats** — *“account suspended”*
3. **Authority** — *“IT department”*
4. **Brands** — *PayPal, Microsoft, IRS*
5. **Credentials** — *“verify password”*
6. **Actions** — *“click here”*
7. **Emotion** — *“congratulations!”*
8. **Money** — *“wire transfer”*
9. **Social Proof** — *“millions of users”*
10. **Attachments** — regex patterns

Why rule-based?

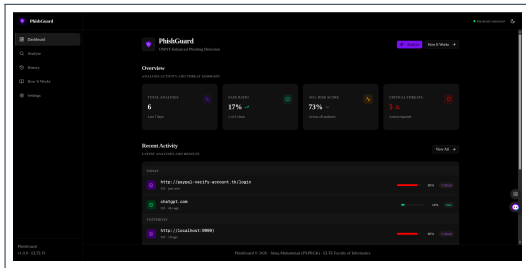
- **Deterministic** — same input, same output
- **Explainable** — exact text evidence
- **Fast** — CPU, milliseconds
- **Maintainable** — add phrases, no retrain

Scoring Weights

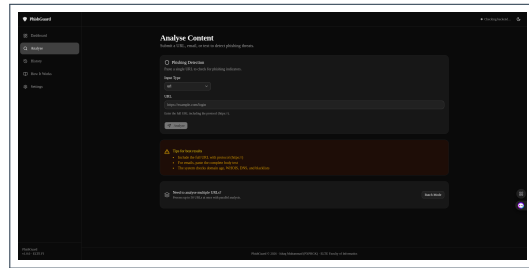
Mode	Primary	Secondary
URL	ML: 85%	NLP: 15%
Email	NLP: 55%	URL 25% + OSINT 20%

The Application

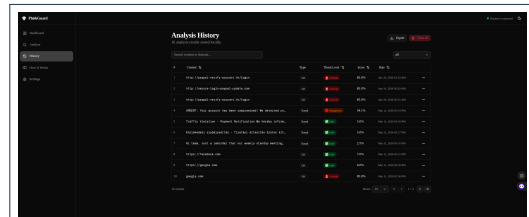
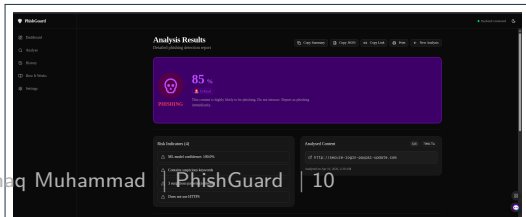
Live Demo Screens



Dashboard



Analyze — URL / Email / Text



Full Tech Stack & Deployment

Backend

- **FastAPI** — async Python REST API
- **XGBoost** — gradient boosting classifier
- **spaCy 3.7** — NLP engine
- **WHOIS + DNS + Reputation** — OSINT
- Deployed on **Render**

Testing

- **592** pytest (backend)
- **133** Jest (frontend)
- **28** Playwright E2E

Frontend

- **Next.js 16** — React framework
- **Tailwind CSS v4 + shadcn/ui**
- **Recharts** — data visualization
- **TanStack Table** — history table
- Deployed on **Vercel**

Live URLs

`project-4soy4.vercel.app`

`phishguard-api-up12.onrender.com`

`github.com/ishaq2321/phishing-detection-osin`

Contributions

1. OSINT-Enhanced Feature Engineering

4 live infrastructure features fused into the ML vector — 14.36% SHAP contribution

2. High-Performance XGBoost Pipeline

96.45% accuracy, 99.41% AUC — Optuna-optimized with 50 trials

3. SHAP Explainability Layer

Every prediction comes with per-feature contribution scores and human-readable reasons

4. Multi-Modal Analysis Engine

Auto-detects URL / email / text — routes through ML or NLP pipeline dynamically

5. Production Full-Stack Application

Next.js + FastAPI, 725 tests, live cloud deployment, Swagger docs

Future Work

Algorithmic

- Online learning for **concept drift**
- Integration of **LLMs** for deeper semantic analysis
- **OCR** for image-based phishing

Engineering

- **Browser extension** for real-time protection
- **CI/CD pipeline** with automated test gates
- **Database** persistence (PostgreSQL)
- **Docker** containerization

Thank You

Questions & Discussion

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github.com/ishaq2321/phishing-detection-osint